In [3]:
```
for i in range(6):
    print(i)
```

```
0
1
2
3
4
5
```

In [15]:
```
def square_numbers(numbers):
    return [num ** 2 for num in numbers]
nums=[2,4,6,8,10]
#nums=int(input("enter a number:"))
squared=square_numbers(nums)
print(squared)
```

```
[4, 16, 36, 64, 100]
```

In [10]:
```
x=[1,2,3,4,5]
total=0
for i in x:
    total=total+i
mean=total/len(x)
print(mean)
```

```
3.0
```

In [17]:
```
import numpy as np
x=[1,2,3,4,5]
mean=np.mean(x)
print(mean)
```

```
3.0
```

In [12]:
```
import numpy as np
x=[1,2,3,4,5]
result=np.min(x)
print(result)
```

```
1
```

In [13]:
```
import numpy as np
x=[1,2,3,4,5]
result=np.max(x)
print(result)
```

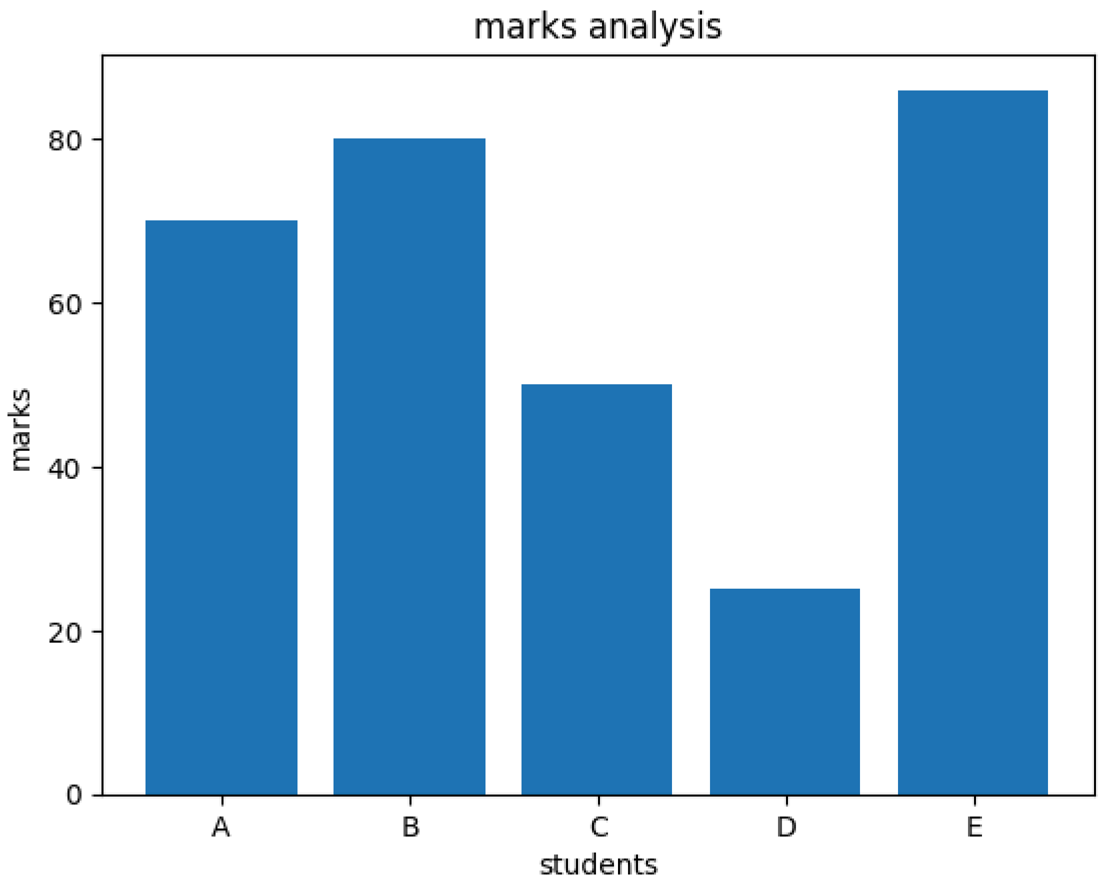
```
5
```

In [18]:
```
variance=np.var(x)
std_dev=np.std(x)
print("Mean:",mean)
print("variance:",variance)
print("standard deviation:",std_dev)
```

```
Mean: 3.0
variance: 2.0
standard deviation: 1.4142135623730951
```

In [20]:
```
import matplotlib.pyplot as plt
stu=["A","B","C","D","E"]
```

```
marks=[70,80,50,25,86]
plt.bar(stu,marks)
plt.xlabel("students")
plt.ylabel("marks")
plt.title("marks analysis")
plt.show()
```



In [ ]: